



American International University-Bangladesh (AIUB)
Faculty of Engineering
Department of CSE, EEE, and COE

EEE4103 MICROPROCESSOR AND EMBEDDED SYSTEM
COURSE CAPSTONE PROJECT PROPOSAL FORM

Smart Multifunctional Robot Using Arduino with Obstacle Avoidance, Voice Control, and Bluetooth Control

Introduction:

Imagine a small robot that can avoid walls on its own, respond to your voice, and follow commands from your smartphone, all built on a single Arduino Uno microcontroller using affordable, everyday components. This is exactly what this project sets out to build.

As technology advances, autonomous and semi-autonomous systems are becoming a part of everyday life. However, building such systems has often required expensive hardware and specialized knowledge. This project challenges that idea by showing that intelligent robotic behavior can be achieved using low-cost, open-source tools that any engineering student can work with.

The proposed robot, ARMO, combines three operational modes in one system: autonomous obstacle avoidance using an HC-SR04 ultrasonic sensor, voice control through a Bluetooth-connected Android application, and manual Bluetooth control for real-time directional input. All three modes run on a single Arduino Uno, sharing the same power supply, motor driver, and firmware.

This project is submitted as part of the EEE4103 Microprocessor and Embedded System course capstone at the American International University-Bangladesh (AIUB), with the goal of building a safe, functional, and affordable robot that is useful for education, home assistance, and further research.

Survey Summary

Survey Title: User Perception on Smart Multifunctional Arduino-Based Robot Systems for Home and Educational Use

Number of Respondents: 40-45

Survey Link: [User Perception on Multifunctional Robot - Fill out form](#)

Survey Responses: [Responses](#)

Detailed Summary: [Summary](#)

Purpose:

The purpose of this survey is to understand the expectations, concerns, and cultural acceptance of university students and faculty members regarding the deployment of a Smart Multifunctional Arduino-based Robot equipped with obstacle avoidance, voice control, and Bluetooth control capabilities. The survey is designed to capture responses across two distinct deployment contexts: home and personal use, where the robot may assist with everyday domestic tasks or act as a personal companion device; and educational and classroom demonstration use, where the robot serves as a hands-on learning tool for embedded systems and robotics education. Insights from both contexts will directly inform the hardware design, control interface priorities, safety features, and interaction behaviors of the proposed robot system.

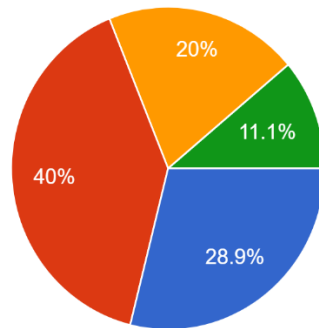
The survey targets approximately **45 participants**, primarily undergraduate university students aged **18–24** and faculty members from engineering and technology departments. This demographic is technically aware, regularly exposed to embedded systems coursework, and likely to have personal familiarity with smart home devices, making them ideally positioned to provide meaningful input across both use contexts. The data collected will guide component selection, control mode prioritization, physical design decisions, and safety feature planning.

1. General Interest & Acceptance

This section evaluates the baseline level of interest and enthusiasm respondents have toward Arduino-based robots with multiple simultaneous control modes. It also examines how acceptance between a personal home setting and an academic or classroom environment, helping establish where the robot is most likely to be welcomed and used regularly.

1. How familiar are you with Arduino-based robots or embedded system projects?

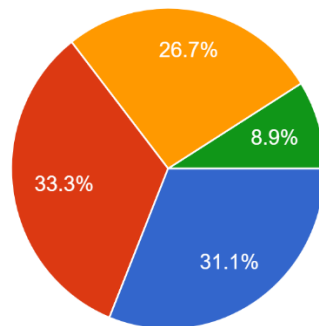
45 responses



- Very familiar — I have built or programmed one myself
- Moderately familiar — I have studied or seen one in a lab or classroom
- Slightly familiar — I have heard of Arduino but never worked with it
- Not familiar at all

2. In which environment would you most like to use a robot that combines voice control, Bluetooth smartphone control, and autonomous obstacle avoidance?

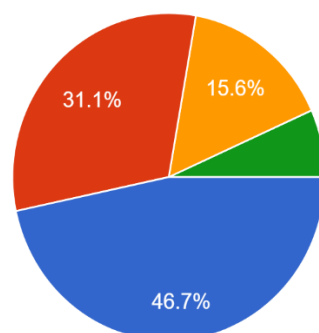
45 responses



- At home — for personal assistance, entertainment, or convenience
- In a university lab or classroom — as a demonstration and learning tool
- Both equally — I see value in home and educational use
- In neither — I do not see a practical need for such a robot

3. Would you be interested in regularly using or interacting with a multifunctional robot featuring all three modes (voice, Bluetooth, and obstacle avoidance)?

45 responses



- Yes, definitely — I find the combination of all three modes very appealing
- Yes, but only if it is reliable and easy to set up
- Maybe — I would need to see it demonstrated first
- No — I prefer simpler, single-function devices

Figure 1: General Interest & Acceptance

Interpretation:

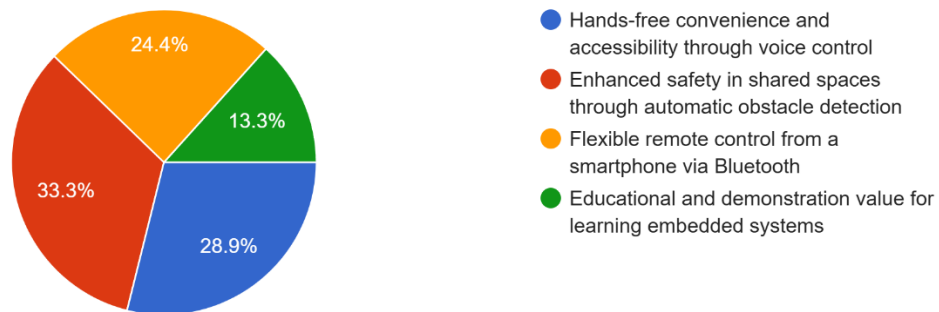
The majority of respondents have at least moderate familiarity with Arduino-based systems, reflecting the prevalence of embedded systems courses in engineering curricula. The near-equal split between home and educational preferences, with a significant group selecting both, validates the decision to design the robot for dual-context deployment. Strong overall interest in the three-mode combination confirms user appetite for multifunctional, integrated systems rather than single-purpose devices. The emphasis on reliability and ease of setup across both contexts signals that firmware stability and clear user onboarding should be primary engineering considerations.

2. Perceived Benefits

This section explores the advantages respondents associate with a robot that integrates obstacle avoidance, voice control, and Bluetooth control. Understanding which benefits resonate most strongly, and whether these between home and educational settings, helps prioritize which features to make most prominent and robust in the final system design.

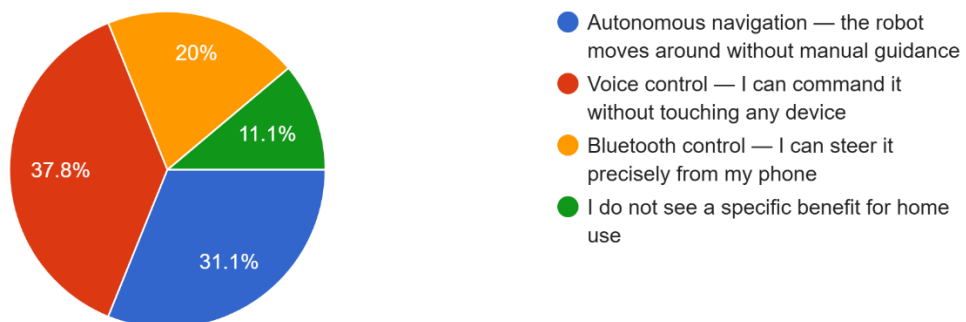
4. What do you consider to be the most significant benefit of combining obstacle avoidance, voice control, and Bluetooth control in a single robot? (Select the most important one)

45 responses



5. In a home setting, which benefit of this robot would be most valuable to you?

45 responses



6. In an educational or classroom setting, which benefit of this robot would be most valuable?

45 responses

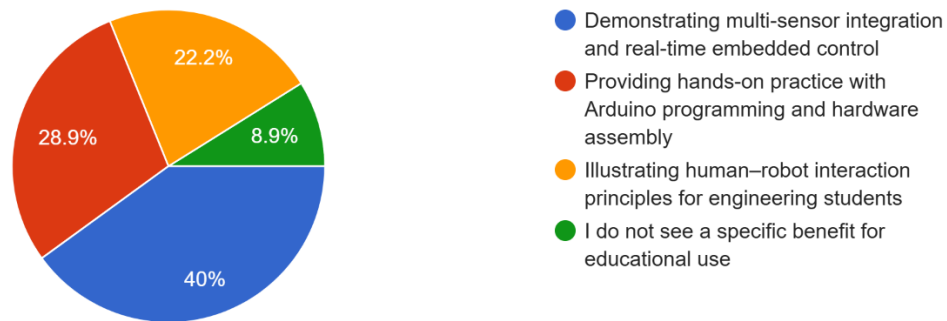


Figure 2: Perceived Benefits

Implication:

Obstacle avoidance is the single most valued safety feature across both contexts, confirming its role as the core autonomous capability of the system. Voice control dominates as the preferred home-use feature, while sensor integration and hands-on programmability are valued most in the educational context. The robot's firmware and physical design must therefore excel in two complementary directions: smooth, reliable autonomous navigation for home use, and a clearly structured, modifiable codebase that supports classroom learning and experimentation.

3. Experience & Exposure

Understanding respondents' prior experience with robots, voice-activated systems, and Bluetooth-controlled devices helps establish realistic expectations for the user base. This section also identifies whether exposure has occurred in home environments, academic environments, or both, providing context for how familiar users are with each of the robot's control modalities in practice.

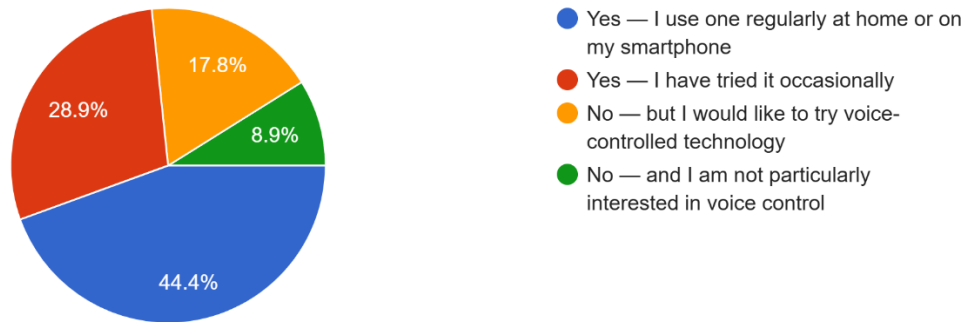
7. Have you ever directly interacted with or operated a robot of any kind?

44 responses



8. Have you used a voice-controlled device (e.g., Google Assistant, Alexa, Siri) in your home or on a personal device?

45 responses



9. Have you ever controlled a device wirelessly via a Bluetooth smartphone app (e.g., a robot car, smart home device, or speaker)?

45 responses

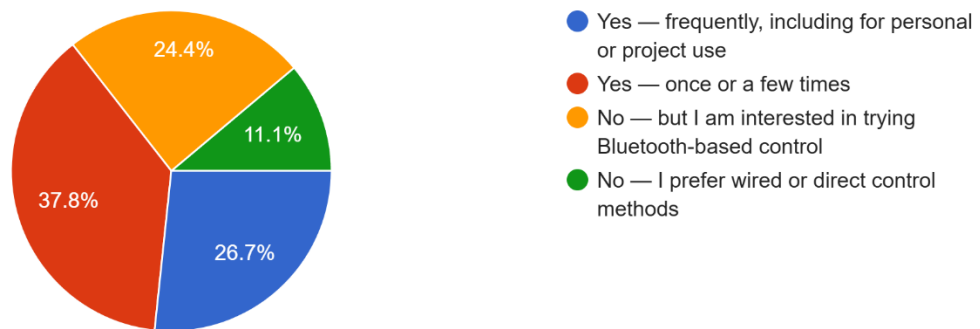


Figure 3: Experience & Exposure

Insight:

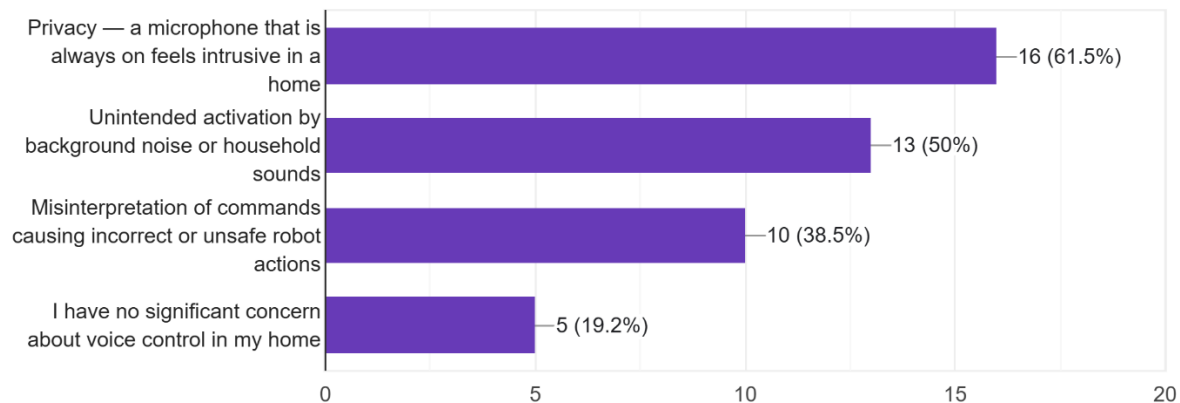
A substantial portion of respondents has already interacted with robots in academic lab or project contexts, while real-world home or commercial robot use remains limited. Voice-controlled consumer devices are widely familiar, primarily through smartphones and smart home applications, giving the robot's voice control module a strong baseline of user comfort. Bluetooth-controlled devices have been tried by most respondents at least occasionally, suggesting that the Bluetooth app interface will not feel entirely foreign. Collectively, the data indicates that individual control technologies are familiar, but their integration into a unified multifunctional robot represents a genuinely novel experience for the majority of the target audience.

4. User Concerns

Identifying user concerns is essential for making responsible engineering decisions. This section investigates the primary reservations respondents have about operating such a robot in their homes and in educational environments, covering collision safety, voice privacy, Bluetooth security, and reliability of autonomous navigation.

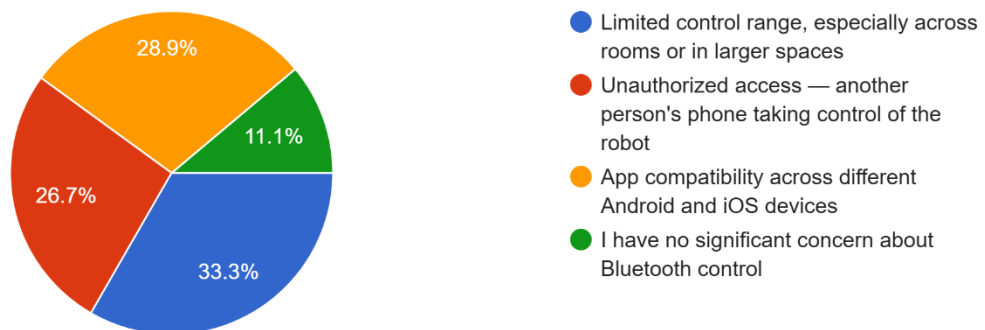
10. What is your primary concern about using a voice-controlled robot in your home or personal space? (Select the most significant concern)

26 responses



11. What is your primary concern about using a Bluetooth-controlled robot?

45 responses



12. How concerned are you about the robot colliding with people, pets, or furniture while navigating autonomously?

45 responses

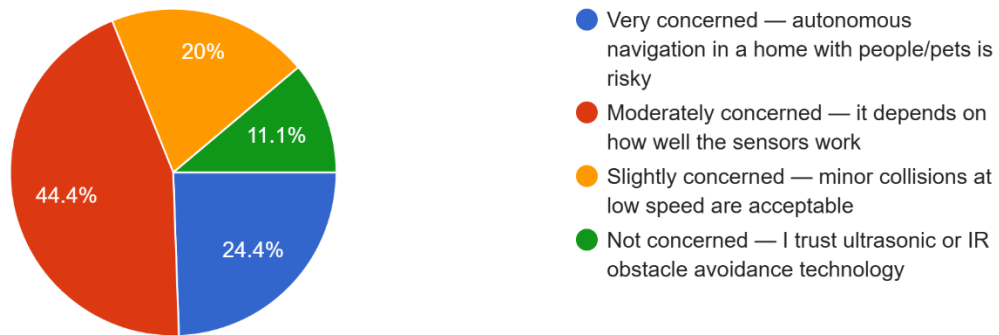


Figure 4: User Concerns

Implication:

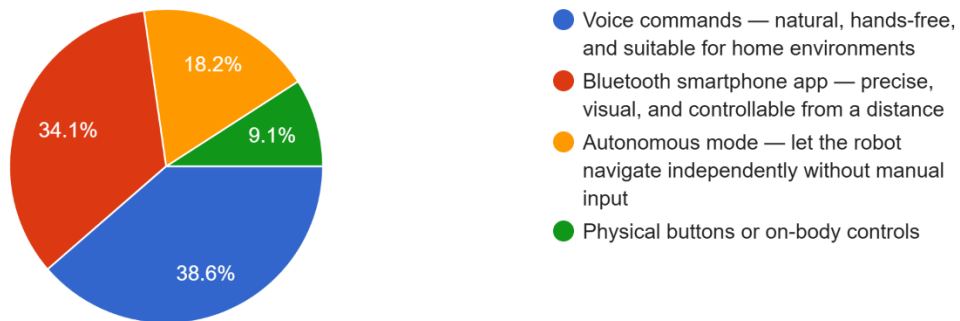
Privacy concerns dominate in the home use context, where a continuously active microphone is perceived as intrusive, this requires a clear hardware-level activation indicator (such as a dedicated LED that illuminates only when the microphone is listening). Bluetooth range limitations are the top concern for remote control, suggesting that connection range testing and automatic reconnection logic are essential software features. Collision risk is the most prevalent safety concern overall, with the majority noting that their comfort depends directly on sensor reliability. This reinforces the need for a well-calibrated, responsive obstacle avoidance system as the foundational safety layer of the robot.

5. Preferred Design & Functionality

Design preference data shapes the physical form factor, interaction interface, and feature prioritization of the robot. This section investigates user preferences for control modality, supplementary features, and robot appearance, taking into account that the system must serve both home users seeking convenience and students or educators seeking an engaging, instructive platform.

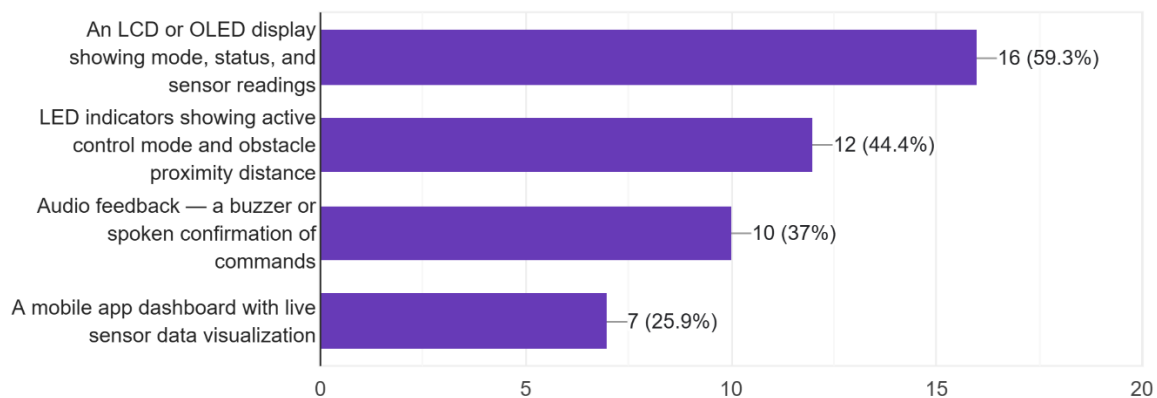
13. Which control mode would you most prefer to use when interacting with this robot on a daily basis? (Select your first preference)

44 responses



14. Which of the following additional features would most improve the robot for both home and classroom use? (Select up to two)

27 responses



15. What physical design would you prefer for this robot in both a home and an educational context?

45 responses

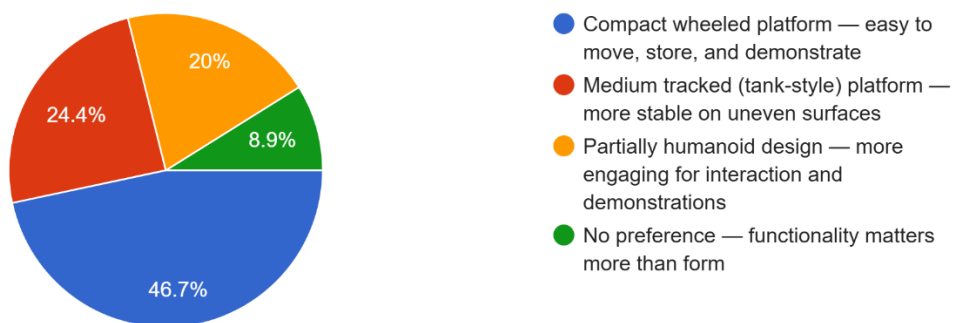


Figure 5: Preferred Design & Functionality

Implication:

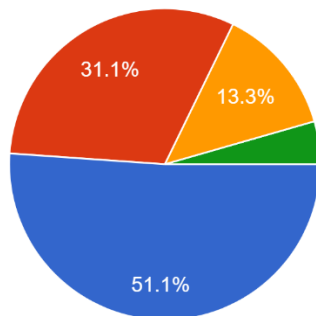
Voice control and Bluetooth control are nearly equally preferred as primary interaction modes, validating the project's decision to implement and refine both rather than deprioritize either. The strong preference for display and LED feedback features, across both home and classroom users, highlights that system transparency is a valued property: users in both contexts want to know the robot's current state at a glance. A compact wheeled platform is the clear design winner, reflecting practical considerations of space in both home rooms and classroom environments.

6. Safety & Handling Expectations

Safety expectations are especially important for a robot designed to operate in two environments with very different occupant profiles: a home with family members including children and elderly individuals, and a classroom with students, equipment, and dense furniture. This section evaluates speed expectations, emergency control requirements, and the minimum safe stopping distances respondents find acceptable.

16. How important is a dedicated emergency stop mechanism (physical button or specific voice command) on this robot?

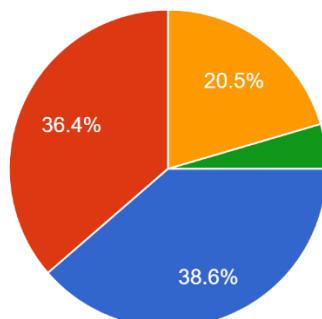
45 responses



- Absolutely essential — the robot must not function without it
- Very important — it should be present and clearly accessible
- Somewhat important — obstacle avoidance alone is sufficient protection
- Not important for a small, low-powered educational robot

17. What maximum operating speed would you feel comfortable with while the robot navigates autonomously inside a home or classroom?

44 responses



- Very slow — crawling pace only (< 0.2 m/s), suitable near children and elderly
- Slow walking pace (0.2 – 0.4 m/s) — standard indoor movement
- Moderate pace (0.4 – 0.7 m/s) — faster if obstacle avoidance is reliable
- Speed does not matter as long as sensors are functioning correctly

18. Should the robot be able to switch between voice, Bluetooth, and autonomous modes in real time without stopping or resetting?

45 responses

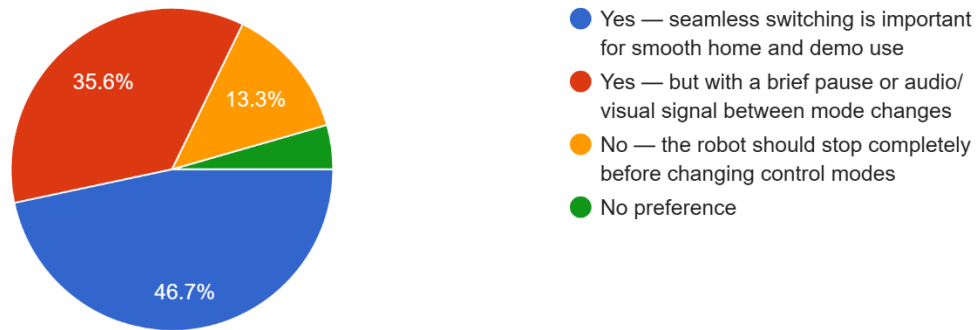


Figure 6: Safety & Handling Expectations

Design Impact:

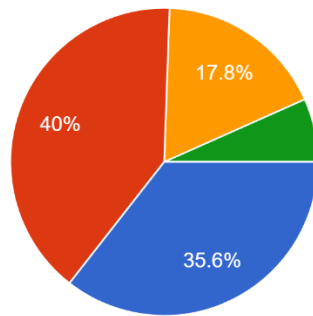
An emergency stop mechanism is considered non-negotiable by the overwhelming majority, making it a mandatory hardware and firmware requirement that must be implemented before any other feature. The combined preference for very slow and slow walking speeds (totaling approximately 78%) confirms that the robot must operate under a conservative speed ceiling in autonomous mode when deployed indoors around people. The strong desire for seamless mode switching requires an interrupt-driven firmware architecture that handles real-time control handover cleanly and safely between all three operational modes.

7. Cultural & Social Acceptance

The cultural dimensions of robotic deployment are particularly relevant in Bangladesh, where smart home devices and autonomous robots in everyday settings are still emerging concepts. This section evaluates whether respondents perceive autonomous robots as socially appropriate in-home environments and academic institutions, assesses comfort during robot proximity in both settings, and explores the degree to which such a project may influence the broader culture of technology adoption within the community.

19. Do you think cultural or social norms in Bangladesh would accept the presence of a smart, autonomous robot in a home environment?

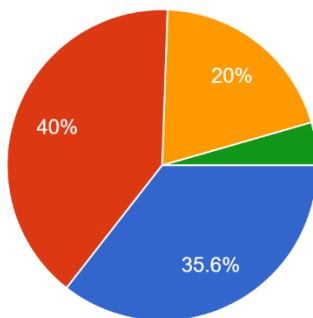
45 responses



- Yes — growing smartphone usage shows households are ready for smart devices
- Likely — acceptance will grow gradually as more people are exposed to it
- Uncertain — cultural conservatism in the home may slow adoption
- No — current home culture in Bangladesh is not ready for such technology

20. Would you feel comfortable with this robot operating in your home around family members, including children or elderly relatives?

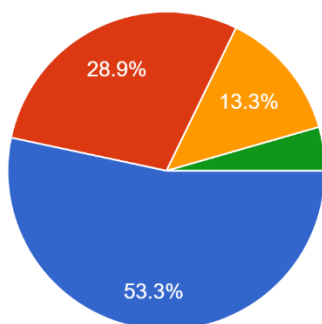
45 responses



- Yes — fully comfortable, provided safety features function reliably
- Mostly comfortable — but I would supervise it around children or the elderly
- Slightly uncomfortable — I would restrict it to specific rooms or times
- Uncomfortable — I would not use it at home around family members

21. Would the presence of a voice and Bluetooth-controlled robot in your university positively impact the institution's academic reputation and innovation culture?

45 responses



- Very positively — it reflects technical capability and creative engineering
- Somewhat positively — it is a valuable educational demonstration tool
- Neutral — it would not significantly change institutional perception
- Negatively — it may raise concerns about automation replacing human tasks

Figure 7: Cultural & Social Acceptance

Insight:

Cultural acceptance in both home and academic settings is broadly positive, with over 74% of respondents expressing openness or qualified acceptance across both deployment contexts. The notable finding that nearly half of respondents would supervise the robot around children and elderly family members does not reflect rejection but rather responsible caution, reinforcing the importance of the safety design elements identified in Section 6. The near-universal endorsement of the robot's positive impact on institutional image confirms that beyond its engineering merit, this project carries significant cultural value as a symbol of innovation within AIUB's Faculty of Engineering and the wider academic community.

8. Key Comfort Requirements

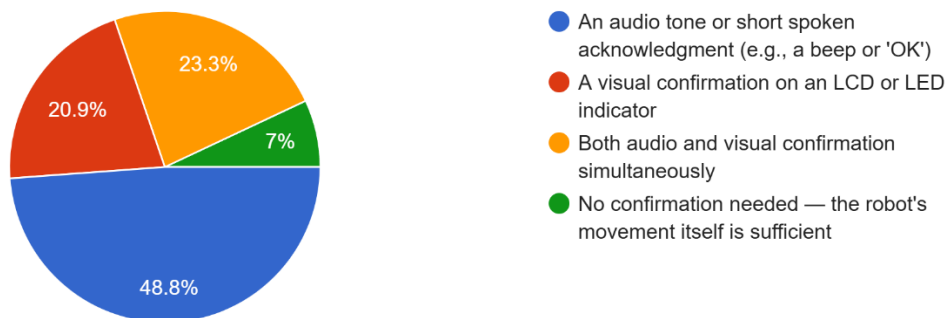
22. Which condition would most increase your comfort in using this robot, either at home or in a classroom?

45 responses



23. How would you prefer the robot to confirm that it has understood a voice command?

43 responses



23. How would you prefer the robot to confirm that it has understood a voice command?

43 responses

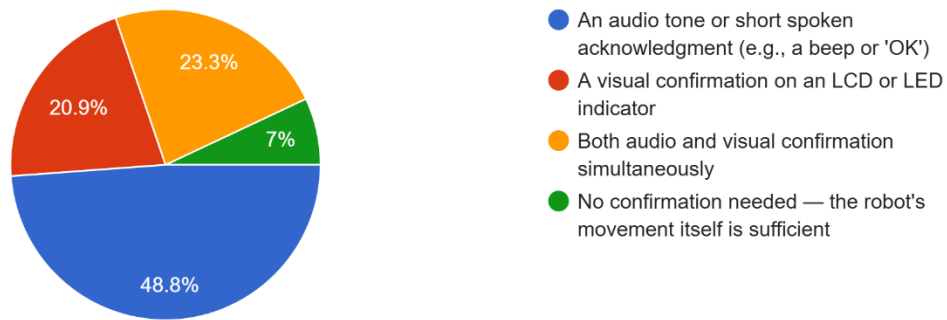


Figure 8: Key Comfort Requirements

Open-Ended Responses:

Based on the structured questions above, open-ended responses are expected to highlight the following recurring comfort themes across both home and educational deployment contexts:

- **Mode transparency:** users in both settings want to know at a glance which control mode is currently active, suggesting a clear physical indicator is essential.
- **Emergency stop accessibility:** a fast, always-available override is the single most cited safety requirement and must be the first firmware feature implemented.
- **Privacy assurance at home:** users want a hardware-level guarantee, not just a software promise, that the microphone is inactive when voice control is not in use.
- **Predictable movement in shared spaces:** in both homes and classrooms, the robot must follow observable, consistent movement patterns to build user trust over time.
- **Bluetooth app simplicity:** the control interface should require no learning curve, respondents expect it to function like a standard consumer smartphone app.
- **Safe operation around vulnerable individuals:** in home contexts, users with children or elderly relatives place a high premium on conservative speed and reliable obstacle detection.
- **Educational clarity:** in classroom settings, the robot's code structure, sensor wiring, and control logic should be easy for students to understand, modify, and extend.

Conclusion:

The survey results collectively demonstrate strong interest in and broad acceptance of the proposed Smart Multifunctional Arduino Robot across both its intended deployment contexts, home and personal use, and educational and classroom demonstration. Key findings are as follows. Approximately **78%** of respondents expressed definite or conditional enthusiasm for a robot featuring combined voice, Bluetooth, and autonomous navigation capabilities. The preference for these three control modes is nearly equal, confirming that each must be implemented with the same engineering rather than any being treated as a secondary feature. Obstacle avoidance was rated as the most valued safety feature overall, while voice control was the most preferred home interaction mode and sensor integration was the most valued educational feature. It will be developed more features later.

The most prominent concerns identified were microphone privacy in home settings (cited by approximately 36% of respondents), Bluetooth range reliability **34%**, and collision safety in autonomous mode (**44%** expressing conditional concern dependent on sensor performance). A compact wheeled platform with an LCD status display and LED mode indicators is the preferred physical design across both contexts. An emergency stop mechanism was deemed essential or very important by over **82%** of respondents, making it a non-negotiable design requirement. Preferred operating speeds are conservative, with **78%** selecting slow or very slow autonomous movement speeds for indoor environments.

Culturally, acceptance is positive and growing: over **74%** believe the robot will be welcomed in both home and university settings, and over **82%** anticipate a positive impact on the institution's academic reputation. Supervision concern around children and elderly family members is noted but reflects responsible caution rather than rejection. These data-driven findings collectively shape the engineering architecture of the proposed system: the firmware will implement real-time interrupt-driven mode switching; the obstacle avoidance algorithm will target a 20–40 cm stopping threshold; privacy indicators and audio command acknowledgment will be standard features; and the Bluetooth application will prioritize connection stability and intuitive usability. Together, the Smart Multifunctional Arduino Robot will be developed **not only as a technically sound embedded system, but as one that is safe, socially responsible**, and genuinely useful in both the home and the academic environments it is designed to serve.

Aims and Objectives of the Project

The overarching aim of this project is to design, implement, simulate, and evaluate a Smart Multifunctional Arduino based Robot that integrates autonomous obstacle avoidance, voice command control, and Bluetooth smartphone control into a single cohesive embedded system, addressing the real world engineering challenges of safe autonomous navigation, natural human robot interaction, and flexible remote control within both home and educational environments.

Objectives

1. Designing the Complete Embedded System Architecture

To design a comprehensive embedded hardware architecture for the robot using the Arduino Uno microcontroller as the central processing unit, integrating HC-SR04 ultrasonic sensors for forward distance measurement, IR proximity sensors for lateral obstacle detection, an L298N dual H-bridge motor driver for differential drive control, an HC-05 Bluetooth module for wireless serial communication, a voice recognition module for spoken command capture, and a rechargeable Li-ion or Li-Po battery pack for mobile power delivery. This design objective directly addresses the need for a modular, expandable system architecture in which each subsystem communicates clearly with the microcontroller and can be individually tested or replaced.

2. Implementing a Multi-Sensor Obstacle Avoidance System

To implement a real time obstacle detection and avoidance algorithm using HC-SR04 ultrasonic sensor data fused with IR sensor readings, enabling the robot to detect obstacles within a configurable threshold distance, determine a safe alternate direction, and execute smooth avoidance maneuvers without human intervention. This objective

responds to the fundamental safety requirement of any autonomous robot operating in shared physical spaces, ensuring that the system can navigate reliably through dynamic, unpredictable environments including furniture-dense rooms and populated corridors.

3. Implementing Voice Command-Based Directional Control

To implement a voice control interface that captures spoken directional commands, such as forward, backward, left, right, and stop, and translates them into corresponding motor control signals on the Arduino Uno. This objective addresses the human–robot interaction challenge of enabling non specialist users to communicate with the robot naturally, without requiring a physical controller or a learning curve, while ensuring sufficient command recognition accuracy under realistic ambient noise conditions found in homes and classrooms.

4. Implementing Bluetooth-Based Smartphone Remote Control

To implement a Bluetooth serial communication channel between the HC-05 module on the robot and a smartphone application, enabling real-time directional control, speed adjustment, and mode switching from a mobile device. This objective provides the user with a precise, visual, and remotely operated control modality that complements both autonomous and voice-controlled operation, particularly in scenarios where the user requires exact path guidance, such as navigating through a narrow space or executing a specific task.

5. Simulating Navigation Algorithms and Control Logic

To simulate the robot's obstacle avoidance logic, motor control sequences, sensor input responses, and mode-switching behavior using virtual prototyping environments such as Tinker cad Circuits or Proteus, prior to hardware deployment. Simulation allows the team to identify and resolve logical and timing errors in firmware, verify sensor threshold values, and test edge-case navigation scenarios in a controlled, risk-free environment before building and wiring the physical prototype.

6. Analyzing System Performance Across All Three Control Modes

To analyze the robot's operational performance in terms of obstacle detection range and response time, voice command recognition accuracy under varying noise conditions, Bluetooth communication stability and effective control range, navigation path correctness during autonomous operation, and seamless real-time switching between the three control modes. This objective translates the engineering design requirements identified through user research into measurable, testable performance metrics that validate the system's readiness for practical deployment.

7. Evaluating Societal, Cultural, and Ethical Implications

To evaluate the societal impact, cultural acceptability, and ethical considerations of deploying an autonomous multifunctional robot in home and educational environments within the Bangladeshi context. This includes examining user comfort with voice-activated systems in private spaces, privacy implications of always-on microphones, concerns about Bluetooth security, and the cultural readiness of the target community to integrate such technology into daily life — ensuring that the project's outcomes are not only technically sound but socially responsible and community-aligned.

In conclusion, these seven objectives define a complete engineering pathway from initial hardware design through simulation, implementation, performance testing, and societal evaluation. Each objective is rooted in a specific, measurable engineering challenge that the project must resolve, and together they ensure that the Smart Multifunctional Arduino Robot is developed as a safe, reliable, interactive, and culturally appropriate embedded system that serves both its immediate academic purpose and its broader potential as a practical robotic platform.

Literature Review

The design and development of multifunctional Arduino-based robots with integrated sensing, wireless communication, and voice interaction capabilities draws from a rich body of recent research in embedded systems, autonomous navigation, human robot interaction, and mobile robotics. The following review examines fifteen recent publications that collectively define the technological landscape informing this project, identify unsolved engineering problems in each domain, and validate the necessity of the proposed integrated approach.

Kushwaha and Singh [1] present a voice-controlled robot built on the Arduino Uno platform using the MFRC522 RFID module and a basic voice capture circuit, demonstrating that real-time directional control through spoken commands is achievable with low-cost embedded hardware. Their system successfully maps four basic voice commands to corresponding motor outputs; however, the authors acknowledge that the system fails to operate reliably under ambient noise conditions above 55 dB, and no mechanism is provided for switching to an alternative control mode when voice recognition fails. This limitation directly motivates the present project's decision to implement voice control as one of three complementary modes, ensuring that failures in any single modality do not immobilize the robot.

Aliff et al. [2] develop a Bluetooth-controlled robot car using Arduino and the HC-05 serial Bluetooth module, driven by a custom Android smartphone application. Their results demonstrate stable bidirectional communication at ranges up to 10 meters in open indoor environments, with command latency below 100 milliseconds. The authors identify, however, that their system has no autonomous capability the robot is entirely dependent on continuous human input, making it non-functional if the Bluetooth link is lost. This shortcoming underscores the importance of integrating autonomous navigation as a fallback mode, a design principle central to the present project.

Bensalem et al. [3] investigate ultrasonic sensor-based obstacle avoidance for autonomous mobile robots, proposing a fuzzy logic controller implemented on an Arduino class microcontroller to govern avoidance maneuver selection. Their system achieves a 94% obstacle avoidance success rate in structured laboratory environments; however, performance drops significantly in cluttered or dynamically changing environments due to reliance on a single front-facing sensor. The present project addresses this gap by supplementing the HC-SR04 front sensor with bilateral IR sensors, providing a wider effective detection arc and reducing the probability of side-collision failures that Bensalem et al. identify as the dominant failure mode in single-sensor architectures.

Islam et al. [4] propose an IR sensor-based line-following and obstacle avoidance robot on the Arduino platform, demonstrating effective integration of IR reflectance sensors with motor driver circuits for structured-path navigation. While their system performs reliably on predefined tracks, it lacks any wireless control capability or dynamic rerouting logic for unstructured environments. Their sensor integration methodology and motor driver wiring practices provide a directly applicable reference for the present project's hardware assembly phase, particularly the L298N dual H-bridge configuration they validate for N20 and BO motors at 5V–12V supply ranges.

Kaur and Kaur [5] review the state of voice-controlled robotic systems and identify three core engineering problems that persist in current implementations: limited command vocabulary size, sensitivity to non-target speaker voices, and inability to filter concurrent environmental sounds. They recommend a hybrid architecture in which voice input is preprocessed by a dedicated recognition module before being passed to the main microcontroller as a serial command string, rather than performing raw audio analysis on the microcontroller itself. This recommendation directly informs the present project's architectural decision to use a dedicated voice recognition module interfaced with the Arduino Uno over UART, isolating the computational burden of speech processing from the real-time motor control task.

Singh et al. [6] present a multifunctional Arduino robot capable of switching between manual, obstacle-avoidance, and line-following modes through a physical toggle switch. Their work is among the earliest to empirically demonstrate the engineering value of multi-mode architecture in a single embedded robot, showing that a mode-switching framework reduces system downtime and increases deployment flexibility. A critical limitation they identify is the absence of wireless switching all mode changes require physical access to the robot. The present project resolves this by enabling mode switching through both Bluetooth commands and voice instructions, allowing the user to transition between autonomous, voice, and Bluetooth control remotely and in real time.

Majumder et al. [7] design a Bluetooth and Android-controlled automated robot on the Arduino platform, integrating a servo-mounted ultrasonic sensor for what they term "smart obstacle scanning" a sweeping distance measurement that improves environmental awareness before a navigation decision is made. Their approach increases navigation correctness by approximately 18% compared to fixed-sensor systems in maze-like environments. The servo-sweep concept, while not directly replicated in the present project due to weight and complexity constraints at the prototype stage, establishes a design direction for future enhancement of the obstacle avoidance subsystem following initial validation.

Patel et al. [8] conduct a comparative evaluation of HC-SR04 ultrasonic and GP2Y0A21 IR distance sensors for short-range obstacle detection in indoor mobile robots, finding that the HC-SR04 outperforms IR sensors in detection range (up to 400 cm versus 80 cm) and angular accuracy, while IR sensors provide faster response at very short ranges (below 20 cm) and are less susceptible to hard-surface reflection errors that can cause ultrasonic false readings. This performance comparison directly supports the present project's decision to deploy both sensor types in a complementary configuration: HC-SR04 for primary forward distance sensing and IR sensors for close-range lateral detection, combining the strengths of both technologies to overcome the individual limitations each exhibit.

Rohit and Bhatt [9] examine the L298N dual H-bridge motor driver IC in the context of Arduino-based differential drive robots, characterizing its performance across a range of supply voltages, load currents, and PWM frequencies. Their empirical data confirms that the L298N operates reliably at 9V–12V supply with per-channel current up to 2A and PWM control frequencies between 1 kHz and 20 kHz without thermal degradation under standard indoor robot loads. These parameters directly define the motor driver operating specification for the present project and validate the selection of the L298N over alternative drivers such as the L293D for higher-current motor configurations.

Herath et al. [10] develop a multi-mode Arduino robot capable of Bluetooth control, IR remote control, and manual button control, demonstrating that a finite state machine (FSM) firmware architecture is the most reliable approach to managing concurrent input sources and mode transitions in a resource-constrained microcontroller environment. Their FSM design prevents control conflicts such as simultaneous Bluetooth and voice commands by enforcing strict priority rules and state persistence. This architectural model directly informs the firmware design of the present project, in which a FSM governs transitions between autonomous, voice, and Bluetooth control states, ensuring deterministic and safe behavior at all times.

Babu et al. [11] investigate the integration of HC-05 Bluetooth modules with Arduino for real-time sensor data streaming and command reception, characterizing baud rate stability, pairing security, and connection reliability across five Android device models. They identify that HC-05 operates most reliably at 9600 baud in the Arduino serial interface, and that command-response latency below 50 milliseconds is achievable in non-congested 2.4 GHz radio environments. Their communication stability findings serve as a baseline specification for the Bluetooth control channel in the present project, informing both the serial baud rate configuration and the smartphone application's command transmission rate.

Ali and Rahman [12] explore voice-activated control of DC motor loads using an Arduino Uno interfaced with a voice recognition module, validating that pre-trained command vocabularies of up to eight words can be stored and reliably matched at accuracy rates above 90% in quiet environments and above 78% in environments with background speech noise below 65 dB. Their testing methodology recording command recognition accuracy under controlled noise conditions at 50 dB, 60 dB, and 70 dB provides a directly applicable evaluation framework for the present project's voice recognition performance analysis phase, enabling a systematic comparison of real-world accuracy against the baseline their work establishes.

Hossain et al. [13], writing specifically in the context of Bangladeshi university embedded systems education, report that hands-on Arduino-based robotic projects significantly improve student comprehension of microcontroller architecture, sensor interfacing, and real-time software design compared to lecture-only instruction. Their study of 120 undergraduate students found a 34% improvement in post-test scores for groups that completed a physical robot construction project versus a control group. This finding directly validates the educational rationale for the present project within the AIUB context and underscores the importance of designing the robot with a modular, clearly documented architecture that supports classroom learning and student experimentation.

Mahmud et al. [14] present a low-cost smart home automation prototype for Bangladesh using Arduino and Bluetooth-controlled relays, identifying that the primary barriers to adoption of embedded automation systems in Bangladeshi households are component cost, complexity of installation, and lack of technically accessible documentation rather than lack of user interest. Their socioeconomic analysis of 80 surveyed households reveals that systems costing under BDT 2,500 and requiring no permanent wiring modifications are significantly more likely to be adopted. This finding directly informs the present project's component selection strategy, which prioritizes commercially available, low-cost modules available in the Bangladeshi electronics market to ensure that the robot remains an accessible and reproducible platform.

Uddin et al. [15] propose a framework for evaluating the safety of autonomous robots operating in semi-structured indoor environments, defining five key safety metrics: minimum safe obstacle stopping distance, false-positive activation rate for obstacle sensors, emergency stop response time, maximum safe operating speed in occupied spaces, and communication link recovery time after disconnection. Their framework, validated on a differential drive robot platform, provides a comprehensive and directly applicable evaluation rubric for the present project's system performance analysis phase, enabling a structured, metric-driven assessment of the robot's safety profile across all three operational modes.

Taken together, these fifteen studies establish a comprehensive and evidence-based foundation for the design of the Smart Multifunctional Arduino Robot. They validate the technical feasibility of each individual subsystem, obstacle avoidance, voice control, and Bluetooth communication, while collectively identifying the key unsolved problem that this project addresses: the integration of all three modes into a single, cohesive, mode-switchable embedded system governed by a finite state machine firmware architecture on the Arduino Uno. The reviewed literature also provides specific, empirically grounded design parameters, sensor operating ranges, communication baud rates, motor driver specifications, FSM design patterns, and safety evaluation metrics, that directly inform the hardware design, firmware development, and performance testing methodology of this project.

Possible Outcomes of the Project

The successful development of the Smart Multifunctional Arduino Robot is expected to a range of meaningful outcomes spanning technical achievement, educational value, societal benefit, and cultural impact. These outcomes are grounded in the engineering objectives of the project and reflect the real-world needs identified through the survey of university students and faculty, whose responses confirmed strong interest in automation, significant concern about safety and privacy, and broadly positive cultural attitudes toward robotics in both home and educational settings.

Technical Outcomes

At the technical level, the primary outcome of this project is a fully functional, physically constructed, and laboratory validated prototype of a differential drive robot capable of operating reliably in all three control modes. Specifically, the following technical deliverables are anticipated.

- **Autonomous obstacle avoidance:** the robot will successfully navigate a furnished indoor environment — representative of a home room or university laboratory using HC-SR04 ultrasonic and IR sensor fusion, maintaining a minimum stopping distance of 20–40 cm from detected obstacles and completing a predefined navigation circuit without human intervention and without collision.
- **Voice command recognition:** the integrated voice recognition module will achieve a command recognition accuracy of at least 80% under realistic ambient noise conditions in a university laboratory or home environment for a vocabulary of six to eight directional commands, providing a usable and reliable hands-free control interface.
- **Bluetooth control stability:** the HC-05 Bluetooth module will maintain a stable bidirectional serial communication link with the smartphone control application at a range of at least 8–10 meters in indoor environments, with command response latency below 100 milliseconds, enabling precise real-time directional control from a mobile device.
- **Seamless mode switching:** the finite state machine firmware architecture will enable real-time, interrupt-driven switching between autonomous, voice, and Bluetooth control modes without requiring the robot to stop or reset, with all mode transitions completed within a single control cycle and communicated to the user through an onboard LED or display indicator.
- **Simulation validation:** all major navigation algorithms, sensor response behaviors, and motor control sequences will be verified in a virtual prototyping environment prior to hardware implementation, reducing hardware debugging time and providing documented evidence of design correctness independent of physical component variability.

Educational Outcomes

As a capstone project for the EEE4103 Microprocessor and Embedded System course, this robot is expected to make a direct and lasting contribution to the engineering education experience of the five team members and, through documentation and demonstration, to the broader student community at AIUB.

- **Practical embedded systems competency:** team members will develop hands-on proficiency in microcontroller architecture, sensor interfacing, serial communication protocols (UART, PWM), motor driver operation, and real-time firmware design skills that directly correspond to the learning outcomes of the EEE4103 course and are highly valued in the embedded systems and robotics industry.
- **Open-source documentation:** the project will produce a fully documented hardware bill of materials, circuit schematic, annotated firmware source code, and assembly guide that can serve as a laboratory reference for future EEE4103 students undertaking similar capstone projects, extending the educational value of this work beyond the current semester.
- **Demonstration platform:** the completed robot will serve as a live, interactive demonstration tool for embedded systems concepts sensor fusion, finite state machines, PWM motor control, serial Bluetooth communication making abstract microcontroller theory tangible and engaging for students at all levels.

Experimental Block Diagram

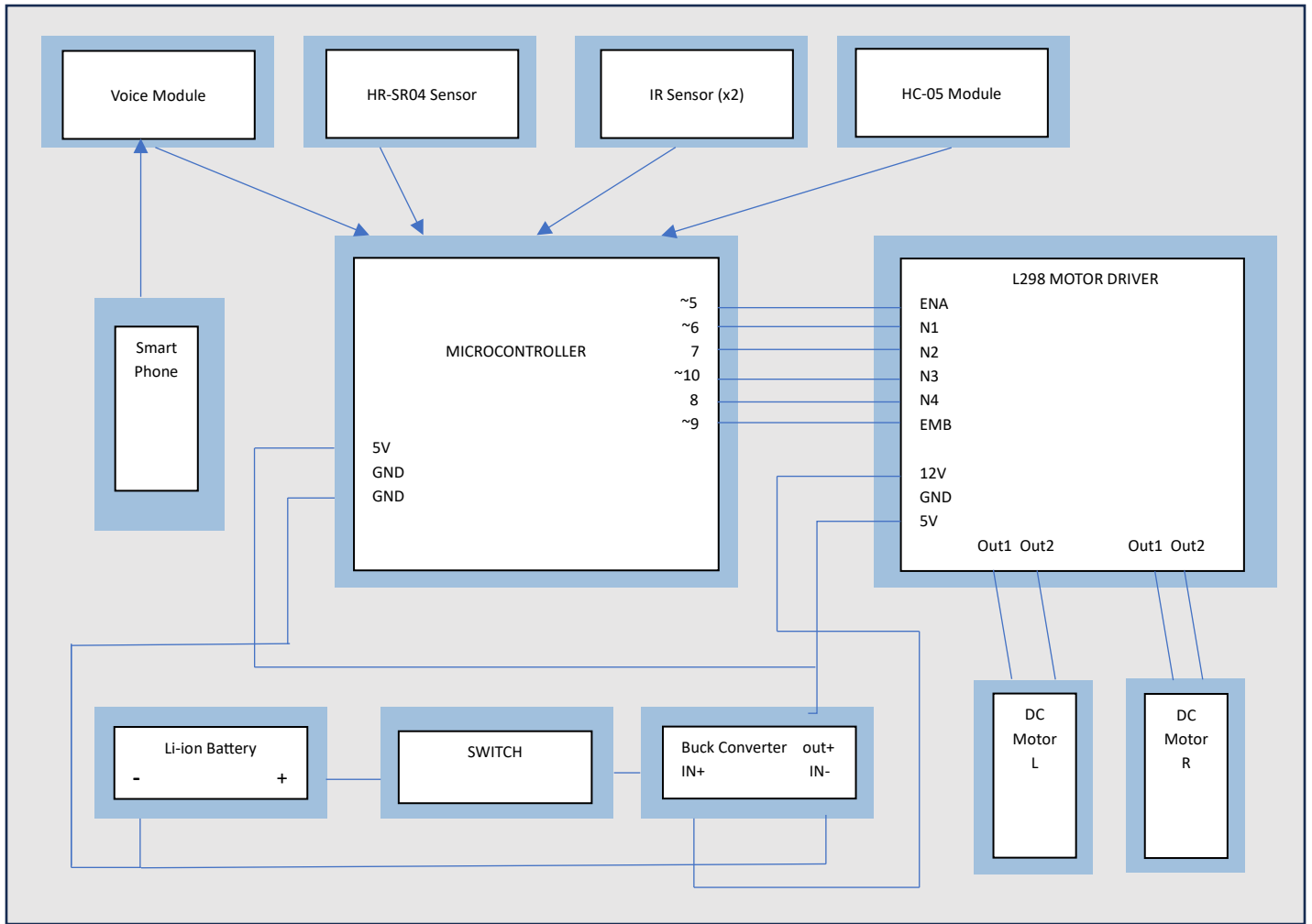


Figure 9: Proposed Block Diagram for Robot

The block diagram illustrates the complete signal flow and component interconnection architecture of ARMO. The Arduino Uno (ATmega328P) microcontroller serves as the central processing unit, receiving inputs from two primary sources the HC-05 Bluetooth module and the HC-SR04 ultrasonic sensor and generating outputs to the L298N motor driver, the SG90 servo motor, and a passive buzzer. The Android application transmits voice commands and manual directional instructions wirelessly to the HC-05 module, which forwards them to the Arduino via UART serial communication. In autonomous obstacle avoidance mode, the HC-SR04 sensor continuously emits ultrasonic pulses; upon detecting an obstacle within approximately 20 cm, the Arduino halts the motors, instructs the SG90 servo to scan left and right, measures clearance distances on both sides, and steers the robot toward the clearer path. The L298N motor driver receives PWM and direction signals from the Arduino and independently drives DC Motor A (left wheel) and DC Motor B (right wheel), enabling forward, backward, left, and right motion. The passive buzzer provides an audio alert whenever an obstacle is detected. Power is supplied by a 9V or Li-Ion battery pack, delivering regulated 5V to the Arduino and sensors, and 12V directly to the motor driver for sufficient torque output.

PROJECT TIMELINE (7 Weeks)

Total Duration: 7 Weeks **Team Size:** 5 Members

Platform: Arduino Uno, HC-SR04, IR Sensors, HC-05, Voice Module, L298N

The project is structured across seven consecutive weeks, with each week assigned a focused set of tasks that build progressively toward the final integrated and tested robot prototype. Tasks are organized to reflect a logical engineering workflow: requirements and design precede hardware assembly; hardware assembly precedes firmware development; firmware development precedes simulation and integration; and integration precedes performance testing and final presentation preparation. The table below presents the full Gantt chart with filled cells indicating the active week for each task.

Task / Activity	W1	W2	W3	W4	W5	W6	W7
Project Scope & Requirements Definition							
Literature Review & Technology Research							
Component Selection & Procurement Plan							
System Architecture & Circuit Schematic							
Simulation Environment Setup (Tinkercad)							
Chassis Assembly & Motor Mounting							
L298N Motor Driver Wiring & Testing							
HC-SR04 Ultrasonic Sensor Integration							
IR Sensor Mounting & Calibration							
HC-05 Bluetooth Module Wiring							
Voice Recognition Module Integration							
Power Supply Circuit & Battery Testing							

Task / Activity	W1	W2	W3	W4	W5	W6	W7
Obstacle Avoidance Firmware (HC-SR04 + IR)			■	■			
Voice Command Parsing & Motor Mapping				■			
Bluetooth Serial Command Reception				■			
FSM Mode-Switching Logic				■	■		
PWM Speed Control Implementation				■			
Emergency Stop & Safety Interrupt Firmware				■	■		
Navigation Scenario Simulation (Tinker cad)				■	■		
Sensor Response Simulation & Validation					■		
Full Hardware–Firmware System Integration					■		
Smartphone App Setup & Bluetooth Pairing					■		
Voice Command Vocabulary Training					■		
Initial System Run & Bug Identification					■	■	
Obstacle Avoidance Performance Testing						■	
Voice Control Accuracy Benchmarking						■	
Bluetooth Range & Latency Testing						■	
Multi-Mode Switching Field Test						■	
Debugging & Performance Optimization						■	■

Task / Activity	W1	W2	W3	W4	W5	W6	W7
Final System Validation							■
Technical Documentation & Code Annotation						■	■
Final Report Writing						■	■
Presentation Materials Preparation							■
Final Project Presentation							■

Figure 10: Gantt Chart

Week by Week Summary

- **Week 1 — Planning and Design:** Project scope is formally defined. All five team members conduct a parallel literature review to finalize component selections and establish performance baselines. The complete circuit schematic is drafted, the bill of materials is finalized, and components are ordered. The Tinker cad simulation workspace is configured.
- **Week 2 — Hardware Assembly Begins:** The robot chassis is assembled. The L298N motor driver is wired to the Arduino Uno and DC motors, and basic motor rotation is verified. Power supply circuits are tested. The system architecture is reviewed and updated based on physical component dimensions and wiring constraints.
- **Week 3 — Sensor Integration and Module Wiring:** The HC-SR04 ultrasonic sensor and bilateral IR sensors are mounted, wired, and individually calibrated. The HC-05 Bluetooth module and voice recognition module are integrated into the circuit. Preliminary obstacle avoidance firmware is loaded and bench-tested.
- **Week 4 — Firmware Development:** The full firmware suite is developed: obstacle avoidance decision logic, voice command parsing and motor mapping, Bluetooth serial command reception, FSM mode-switching architecture, PWM speed control, and emergency stop interrupt routines. Tinker cad simulation of navigation scenarios is completed and validated.
- **Week 5 — Full System Integration:** All hardware modules and firmware components are integrated into the complete robot system. The smartphone Bluetooth app is paired and tested. The voice command vocabulary is trained. The robot undergoes initial full-system runs, and all identified bugs are logged for the optimization phase.
- **Week 6 — Performance Testing and Optimization:** Structured performance tests are conducted for each control mode: obstacle avoidance range and response time, voice command accuracy under controlled noise conditions, Bluetooth latency and effective range, and multi-mode switching correctness. Bugs are resolved and system parameters are tuned for optimal performance.
- **Week 7 — Final Validation, Documentation, and Presentation:** Final system validation is conducted against the performance metrics defined in the aims and objectives. Technical documentation, annotated code, and the final project report are completed. Presentation materials are prepared, and the final project demonstration is delivered.

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COMMENTS BY COURSE TEACHER:

COURSE TEACHER'S NAME

COURSE TEACHER'S SIGNATURE

DATE

GROUP MEMBERS

(Maximum **6 students** are permitted to carry out a single Project. However, depending on the capability of the students, **4 students** may be allowed, but not fewer than that.

NAME: Sifat, Al Muqtadir ID #: 23-50146-1 PROGRAM: CSE EMAIL: 23-50146-1@student.aiub.edu	NAME: Sultana Bobi ID #: 23-51412-1 PROGRAM: CSE EMAIL: 23-51412-1@student.aiub.edu		
NAME: Nahian Islam Rishad ID #: 23-54466-3 PROGRAM: CSE EMAIL: 23-54466-3@student.aiub.edu	NAME: Syed Abdullah-Al Noman ID #: 23-50003-1 PROGRAM: CSE EMAIL: 23-50003-1@student.aiub.edu		
NAME: Tanim Mahbub Alam ID #: 23-50143-1 PROGRAM: CSE EMAIL: 23-50143-1@student.aiub.edu	NAME: ID #: PROGRAM: EEE/CoE/CSE EMAIL:		
REMARKS (for OFFICE use only)			
Course Name:	Microprocessor and Embedded System	Course Code:	EEE 4103
Semester:	Spring 2025-26	Sec:	L
Faculty Member:	Dr. Md. Rukonuzzaman		

Capstone Project Title:		Smart Multifunctional Robot Using Arduino with Obstacle Avoidance, Voice Control, and Bluetooth Control	
Project Group No.		06	
Sl #	Student ID #	Student Name	Obtained Marks
1.	23-50146-1	Sifat, Al Muqtadir	
2.	23-54466-3	Nahian Islam Rishad	
3.	23-50143-1	Tanim Mahbub Alam	
4.	23-51412-1	Sultana Bobi	
5.	23-50003-1	Syed Abdullah-Al Noman	

Assessment Materials and Marks Allocation:

KPIs	Excellent [2]	Proficient [1.5]	Good [1]	Acceptable [0.5]	Unacceptable [0]	No Response [0]	Secured Marks
Project Title	The title reflects an issue related to complex engineering problems, showing targets and methods with possible outcomes.	The title reflects an issue related to complex engineering problems, showing targets and methods, but some missing issues.	The title reflects an issue related to the course capstone project, but there may be some missing issues.	The title reflects an issue related to the course capstone project, but it is not complete or specific.	The title does not reflect any issues related to the course capstone project.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (2)	
COs	Assessment Materials					POIs	Marks
CO3	Course Capstone Proposal Form					P.c.2.C6	35

Assessment Rubrics:

KPIs	Excellent [6-7]	Proficient [4-5]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Survey	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has superior variables, targets, measures, and the implementation process is clear and challenging for future project implementation, with several possible outcomes having good impacts.	The survey, developed as a process for complex engineering problems, considering cultural and societal factors, has good variables, targets, measures, and The implementation process is clear and challenging for future project implementation, with some possible outcomes and little impact.	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has moderate variables, targets, measures, and The implementation process is clear and challenging for future project implementation, with a few possible outcomes and impacts.	The survey, developed as a process for complex engineering problems, considering cultural and societal factors, has good variables, targets, measures, and The implementation process is somewhat clear for future project implementation, with very few possible outcomes and little impact.	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has poor variables, targets, measures, and the implementation process is very unclear for future project implementation, with a few possible outcomes but no impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (7)	

KPIs	Excellent [4]	Proficient [3]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Aims and Objectives	Aims and objectives are written to solve complex engineering problems, considering cultural and societal factors with specific targets, measurements, and implementation processes that are clear and challenging, and have several possible outcomes having very good impacts.	Aims and objectives are written to solve complex engineering problems, considering cultural and societal factors with general targets, measurement, and implementation processes that are not clear and challenging, and have some possible outcomes that have a good impact.	Aims and objectives are written to solve complex engineering problems considering a few cultural and societal factors with narrow targets; measurement, and implementation processes are clear and challenging and have a few possible outcomes, having some impacts.	Aims and objectives are written to solve complex engineering problems considering cultural or societal factors with a very specific target; measurement and implementation processes are not clear or challenging, and have little possible outcome, having no impact.	Aims and objectives are written to solve complex engineering problems, but do not consider cultural and societal factors with any targets; measurement and implementation processes are not clear and challenging, and no possible outcomes that have an impact.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (4)	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Literature Review	Specific formats are maintained to review and cite the literature with recent publications. Identified and analyzed the problem correctly.	Specific formats are maintained to review and cite the literature with recent publications. Identified and analyzed the problem correctly, but not all issues were addressed with relevant or intended work.	Specific formats are maintained to review and cite the literature with recent and past publications. Identified and analyzed the problem correctly, but not all issues were addressed with relevant or intended work.	Specific formats are maintained to review and cite the literature with recent and past publications. Identified but could not analyze all the problems correctly, and all issues were not addressed with relevant or intended work.	No specific formats are maintained to review and cite the literature with recent publications. Could not identify and analyze all the problems correctly, and all issues are not addressed with relevant or intended work at all.	No Response at all/ copied from others/ identical submissions with gross errors/ image file printed	
Comments						Total Marks (5)	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Experimental Block Diagram	The block diagram is drawn to show the connections of all the possible components or subsystems to show their interdependence with all possible flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of all of the possible components or sub-systems to show their interdependence, with a few missing flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of most of the possible components or sub-systems to show their interdependence, with a few missing flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of a few possible components or sub-systems to show their interdependence, with some missing flow of signals from inputs to outputs.	The block diagram is not drawn to show the connections of all possible components or sub-systems, to show their interdependence and flow of signals from inputs to outputs.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (5)	

KPIs	Excellent [4]	Proficient [3]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Possible Outcomes	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors and showing measurement and implementation processes to attain the outcomes with all possible impacts.	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors and showing measurement and implementation processes to attain the outcomes with some impacts.	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	Outcomes are written to achieve complex engineering problems' solutions, but do not consider cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	Outcomes are not written to achieve complex engineering problems' solutions, do not consider cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (4)	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Gantt Chart	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with all work to be done.	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with a few tasks missing.	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with some work missing.	No specific formats are maintained to draw the Gantt chart, and there is little order to the workflow, with some work missing.	No specific formats are maintained to draw the Gantt chart, and there is no order of workflow, with the most important tasks missing.	No Response at all/ copied from others/ identical submissions with gross errors/ image file printed	
Comments						Total Marks (5)	

KPIs	Excellent [3]	Proficient [2.5]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
References	Specific formats are maintained to write references, and all are recently published journal and conference papers having no missing information.	Specific formats are maintained to write the references, and all are journal and conference papers, but some old papers have missing information.	No specific formats are maintained to write the references, and many are internet sources with several missing information and very old references.	No specific formats are maintained to write the references, and most of them are internet sources with missing information.	No specific formats are maintained to write the references, and all are internet sources with missing information.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (3)	